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AC/DC Power Supply

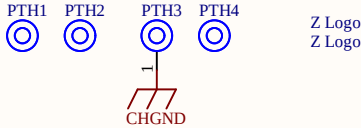
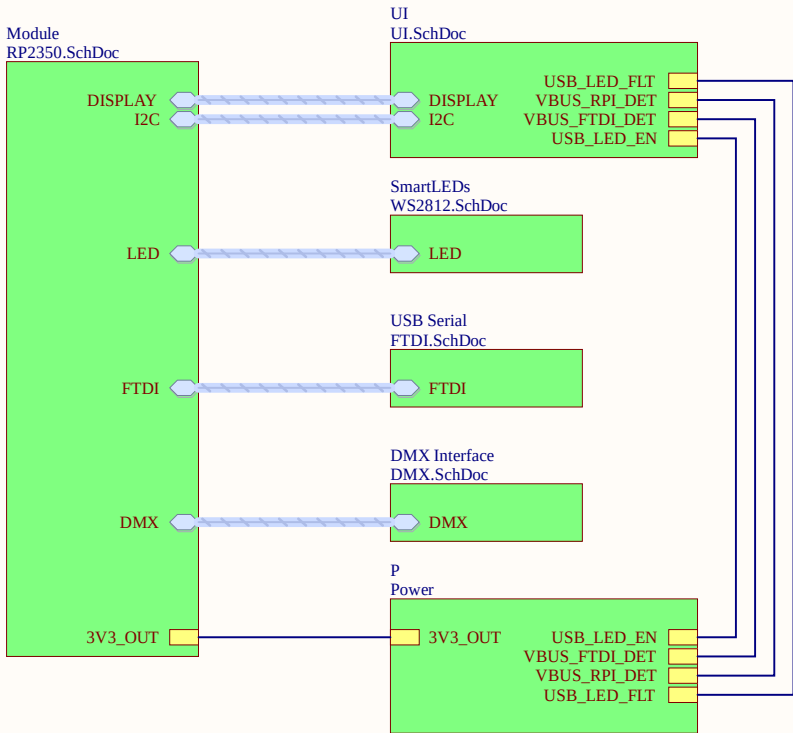
5V 200W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200A-5/7707187>

12V 162W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/LSP-160-12T/11689253>

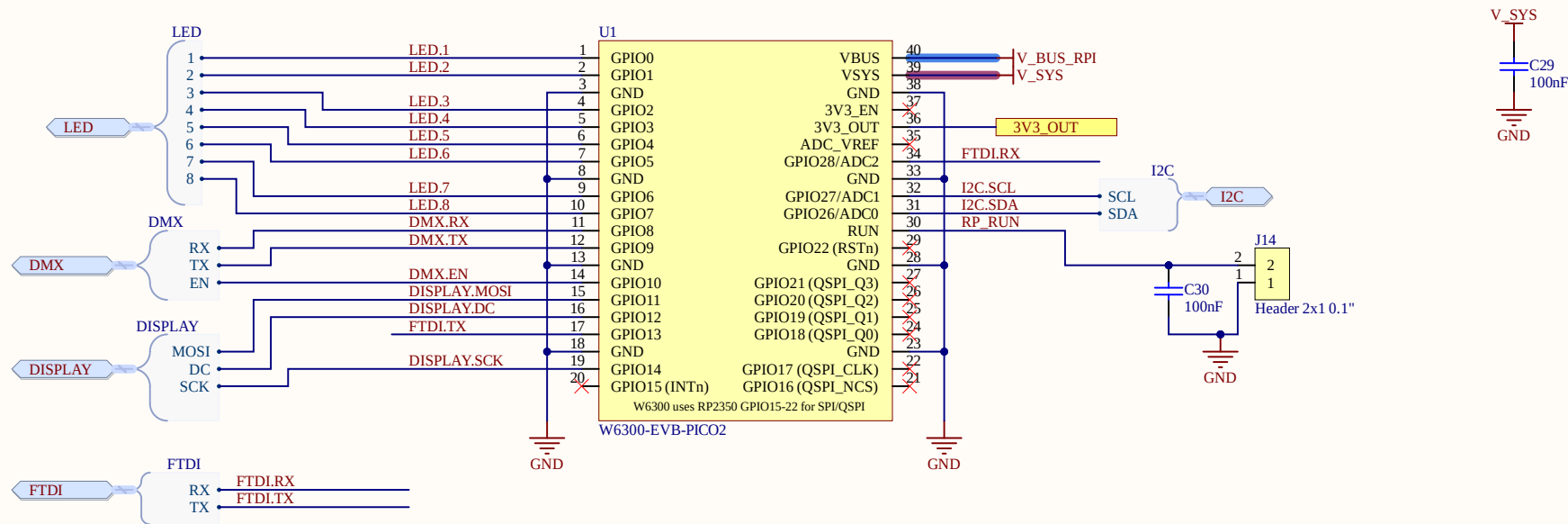
12V 200W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200-12/7707237>

24V 162W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/LSP-160-24T/11689249>

24V 200W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200-24/7707239>



Title			
Block Diagram			
Size	Number		Revision
B	DMX Interface		2
Date:	9/04/2026		Sheet of 1 of 7
File:	C:\Users\...\Block Diagram.SchDoc		Drawn By: *

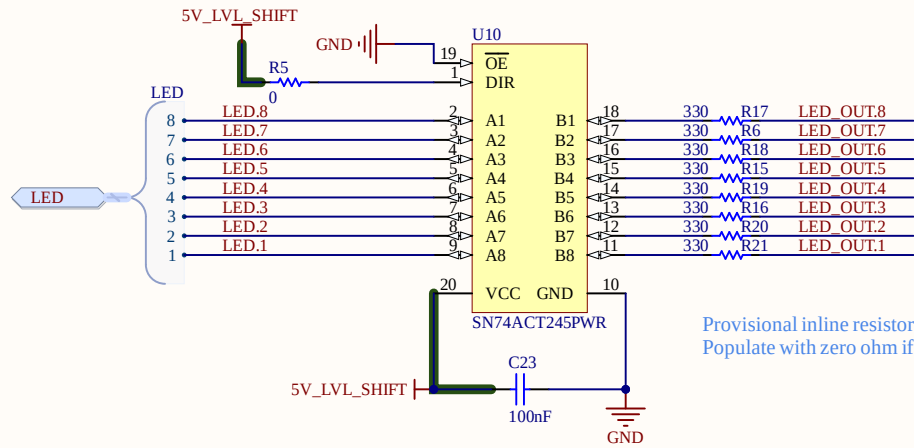


Socket the module on two 1x20 sockets

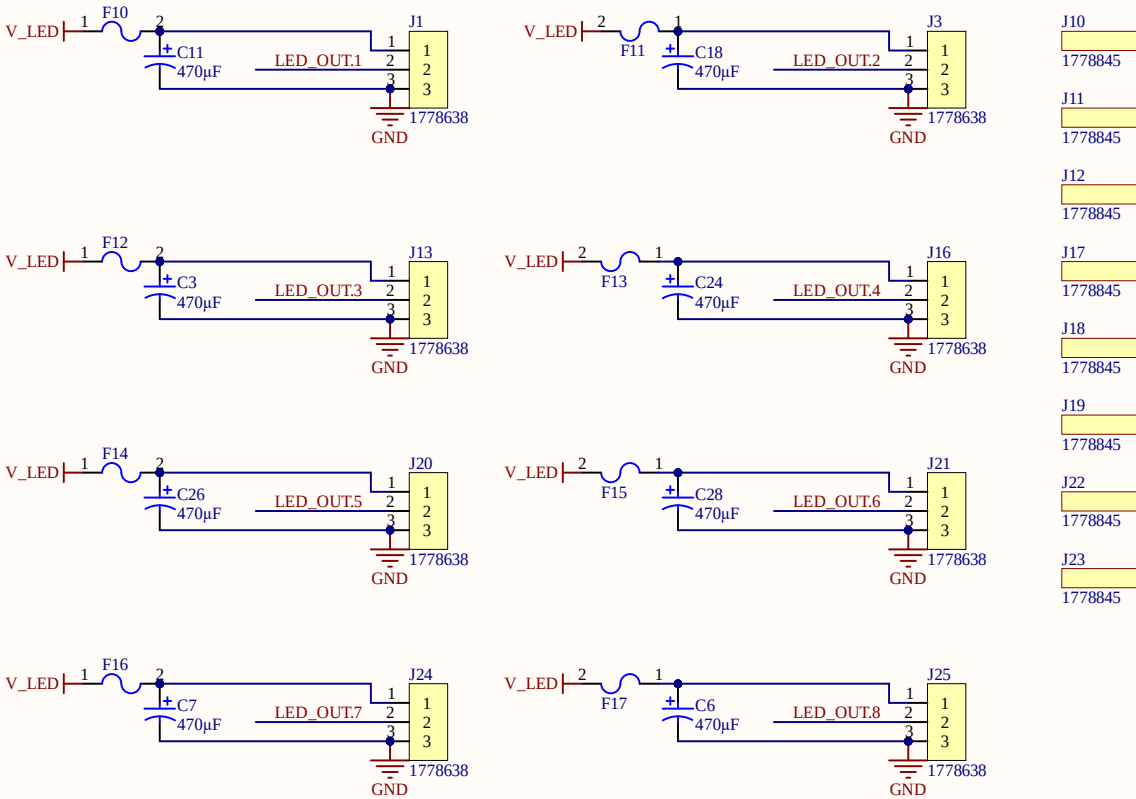
GP15-22 (pins 20-22, 24-27, 29) are hard-wired to the on-module W6300; they are marked no-connect and must stay that way.

3V3_OUT (pin 36) feeds ONLY the carrier LDO - the sequencing gate that keeps carrier 3V3 from rising before IOVDD. No carrier load runs from it.

Title RP2350			
Size B	Number DMX Interface		Revision 2
Date:	9/04/2026	Sheet of	2 of 7
File:	C:\Users\...\RP2350.SchDoc	Drawn By:	*



Provisional inline resistor required for some LED strips.
Populate with zero ohm if not needed.



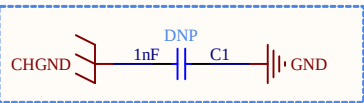
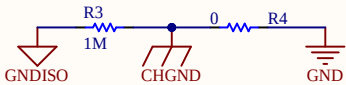
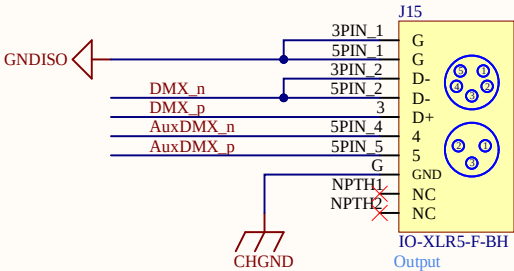
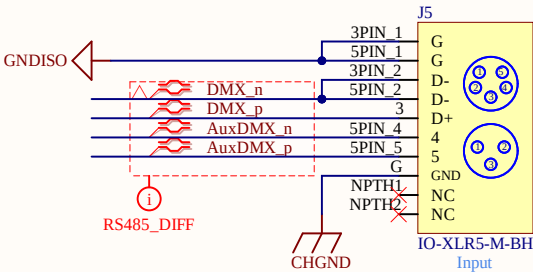
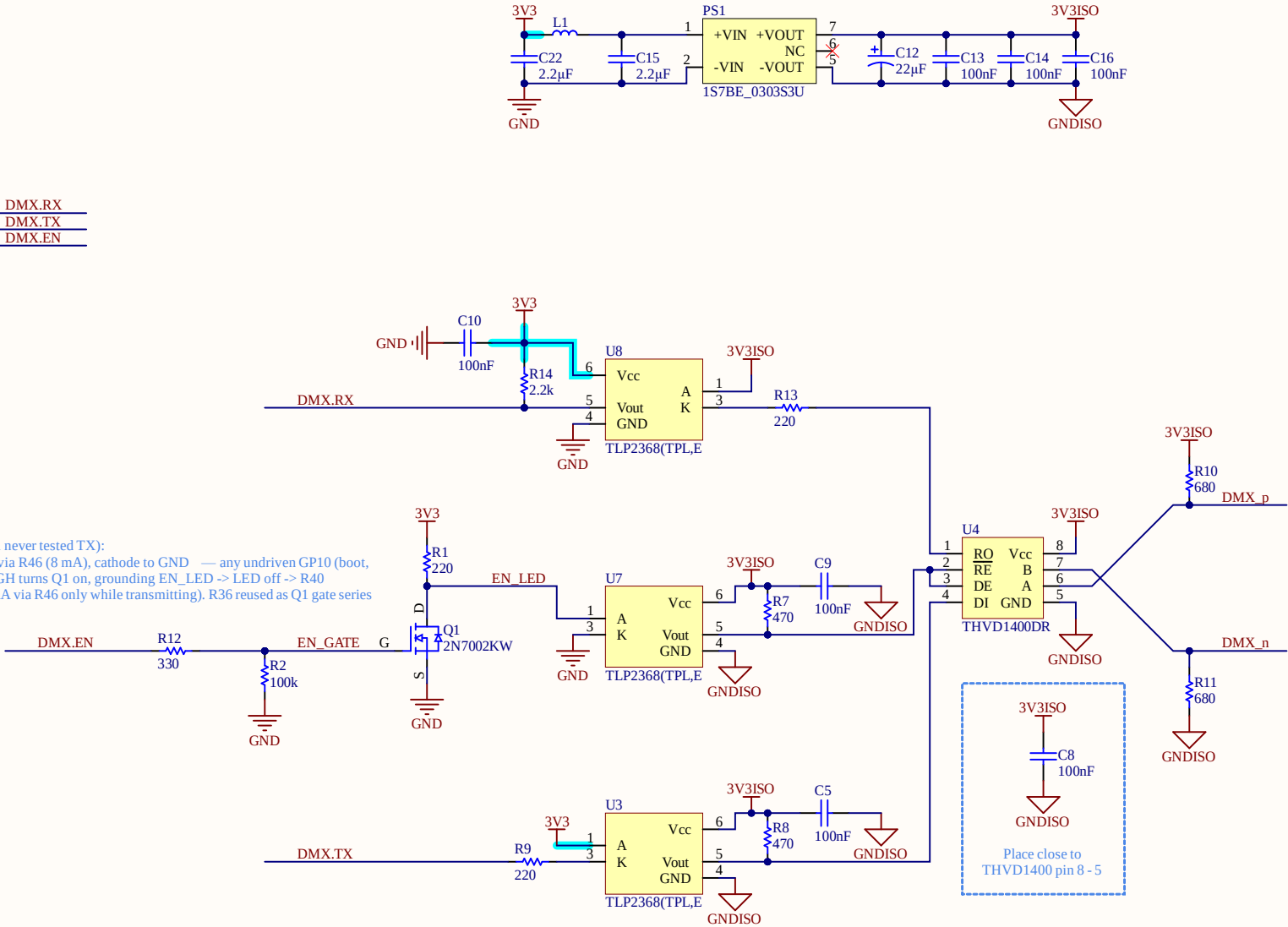
Mating connector: 3 pin 2.5mm pitch : 1778845

24V variant: replace fuses with 0ZCF0150FF2C

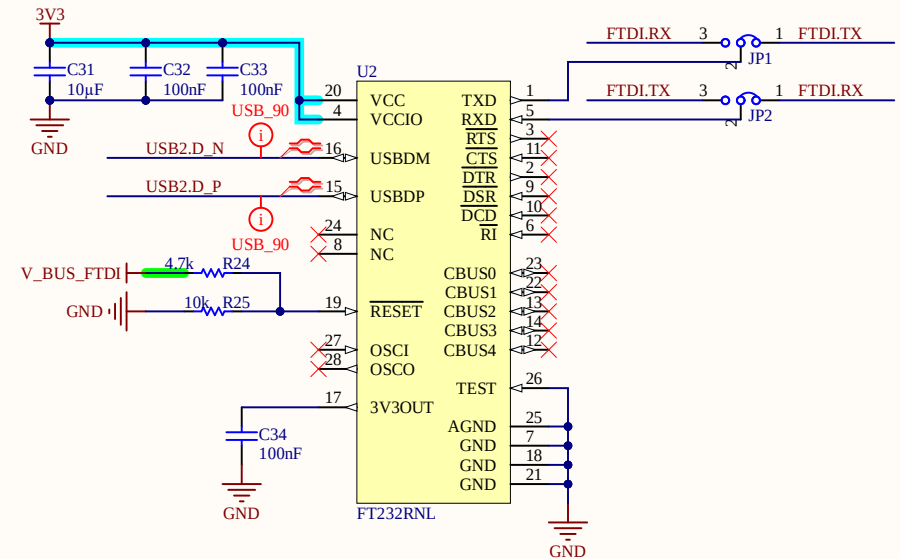
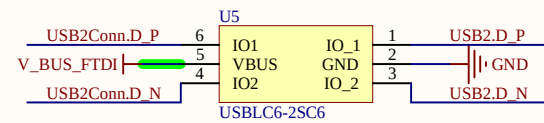
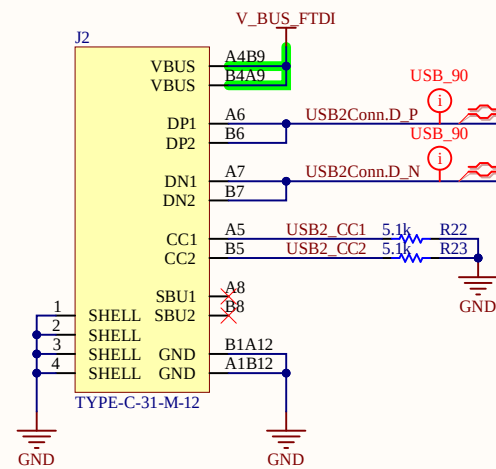
Title			Smart LED		
Size	Number	DMX Interface		Revision	2
B					
Date:	9/04/2026	Sheet of	3	of	7
File:	C:\Users\...\WS2812.SchDoc	Drawn By:	*		



Fail-safe DMX direction (Rev 2 rework — Rev 1 never tested TX):
DMX EN OPTO ISO LED biased ON from 3V3 via R46 (8 mA), cathode to GND — any undriven GP10 (boot, reset, unflashed, crashed) = RECEIVE. GP10 HIGH turns Q1 on, grounding EN_LED -> LED off -> R40 (3V3ISO) pulls DE/RE high = TRANSMIT (15 mA via R46 only while transmitting). R36 reused as Q1 gate series



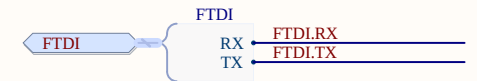
Title DMX / RS485		
Size B	Number DMX Interface	Revision 2
Date: 9/04/2026	Sheet of 4 of 7	
File: C:\Users\...\DMX.SchDoc	Drawn By: *	



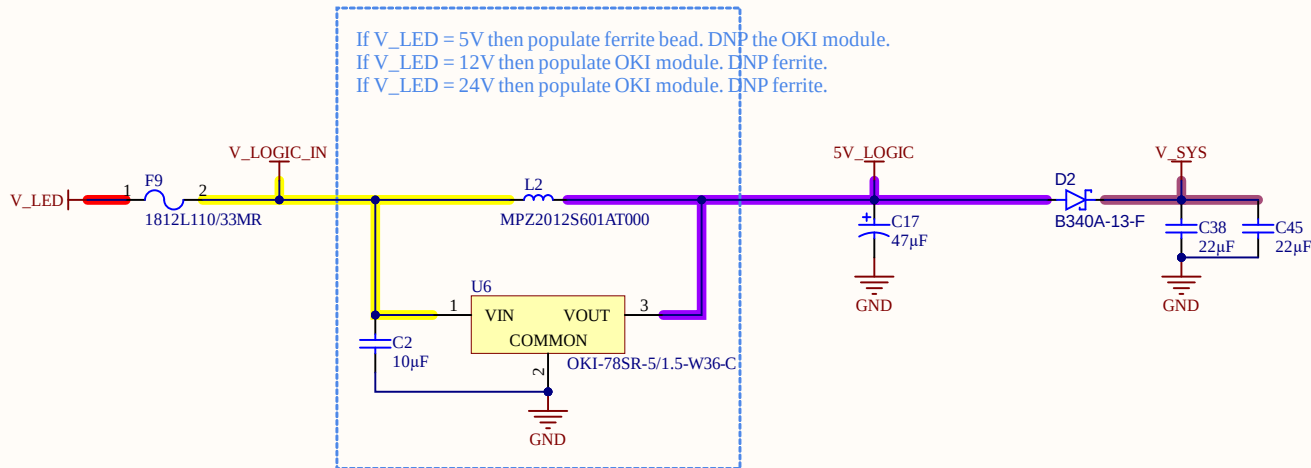
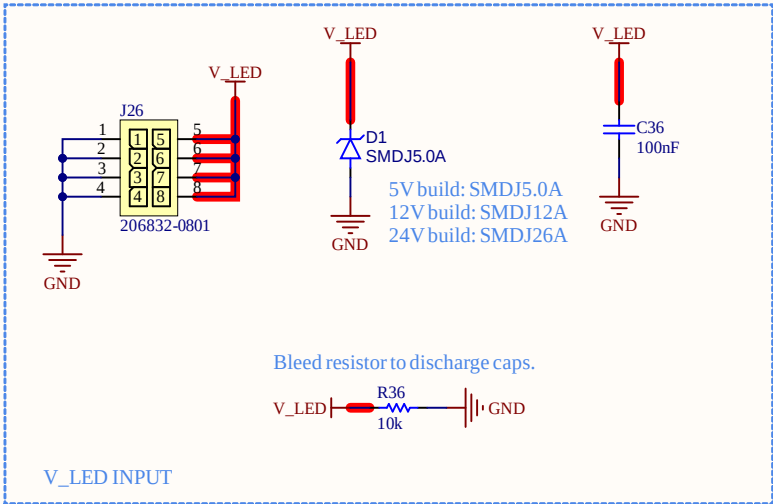
This FT232R is self-powered from the board's always-on 3V3, unlike a normal bus-powered dongle. A running FT232R asserts its internal 1.5 kΩ pull-up on D+ (that's how USB devices announce themselves). If RESET# were simply tied high, then whenever the box is on, the chip would drive ~3.3 V into D+ of whatever it's cabled to — telling a sleeping PC “a device just attached,” back-feeding an unpowered hub port, and generally misbehaving on the far end. Held in reset, D+ stays released until real VBUS proves a host is actually there. This is verbatim FTDI's recommended self-powered configuration from the FT232R datasheet (their “Self Powered Configuration” shows this exact 4.7 k/10 k divider), and it's why the schematic note on the usb_dmx sheet reads “no host, no enumeration, no back-drive.”

The contrast, for completeness: in a bus-powered design (chip's VCC comes from VBUS), you'd just tie RESET# to VCC or leave it floating on its internal pull-up, because the whole chip dies with the cable anyway. Ours doesn't — hence the gate.

One practical bench note: this means the FT232R port only enumerates when both the box is powered and a host is attached — if you ever probe USB2_VBUS at 5 V but FTDI_RST isn't ~3.4 V, check RESET line voltage divider resistors stuffing before anything else.



Title		
Size B	Number	Revision
Date:	9/04/2026	Sheet of
File:	C:\Users\...\FTDI.SchDoc	Drawn By:



INPUT: NO main fuse on the board: a 40-50 A MIDI/ANL fuse in the supply harness at the PSU is REQUIRED — it protects the V_LED plane and is what D1 crowbars against on overvoltage. Per-port protection: 8 x PTC fuses on the WS2812 sheet; F9 guards logic.

V_LED is the strip rail: 5 V OR 12 V per BUILD (one voltage per board — silk-mark it).

3V3 here IS the diagram net 3V3_CARRIER: TLV75733 (dropout 450 mV max at 1 A, ~80 mV at our ~175 mA incl. FT232RNL + TFT backlight) regulates from 3.95 V worst-case USB input. Thermal pad to GND pour. Module 3V3(OUT) feeds nothing on the carrier.

SEQUENCING: U1 EN high = on, FAULT to P07, 18k ~ 1.45A, is gated by the module 3V3_OUT, so the carrier 3V3 rail rises only after the RP2350 IOVDD is up — without this, the I2C pull-ups and opto anodes back-inject ~15 mA into unpowered, non-failsafe RP2350 pads at every power-up.

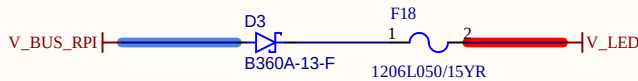
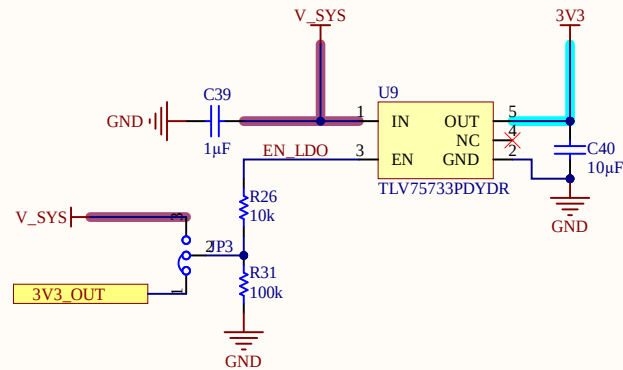
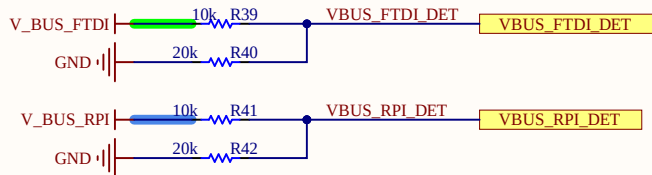
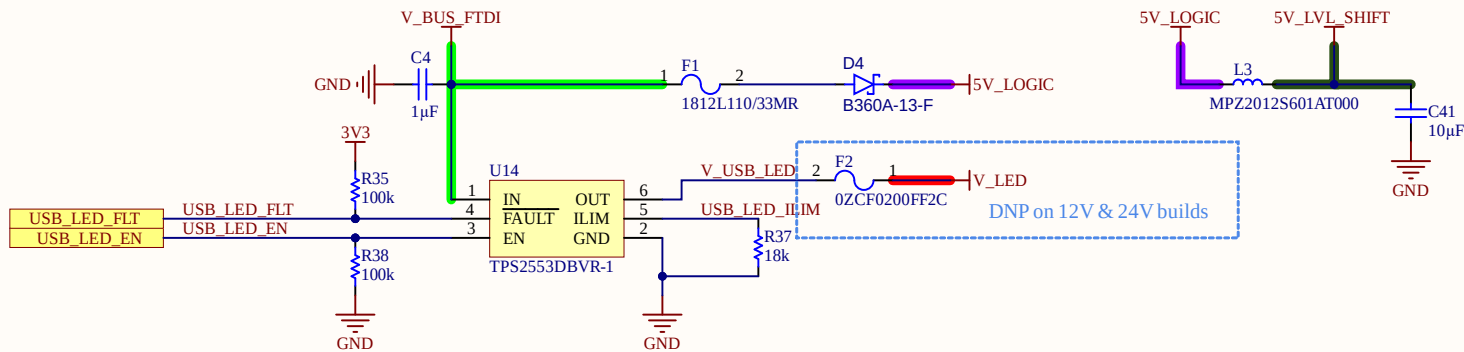
F2 DNP on 12/24 V, PSU trim ≥ 5.2 V.

RATING: 8 ports x 5 A = 40 A worst-case through the board

Layout: 2 oz copper, V_LED and GND as paired pours, strip returns meet logic ground at J1 only.

CHGND bond: R4 fitted / C1 DNP are the tuning points.

THERMAL: D1 (B340A, SMA ~90 K/W on decent pads) dissipates ~200 mW at the ~550 mA logic+TFT load (~+20 C) — give it copper on both pads. VSYS = 5.0 V - Vf ~ 4.6 V, ample for the module buck. Same SMA pad takes onsemi SS34 or MCC SK34.



Title		
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Date:	9/04/2026	Sheet of
File:	C:\Users\...\Power.SchDoc	Drawn By:

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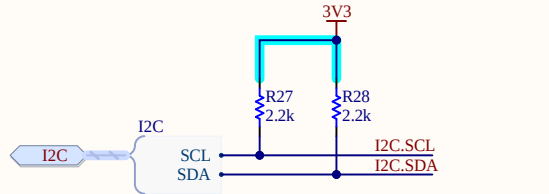
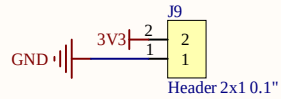
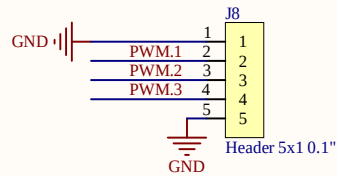
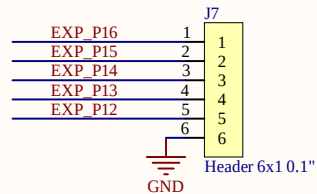
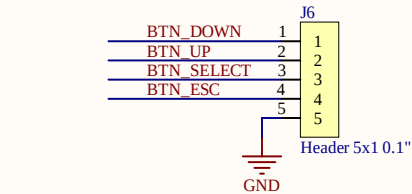
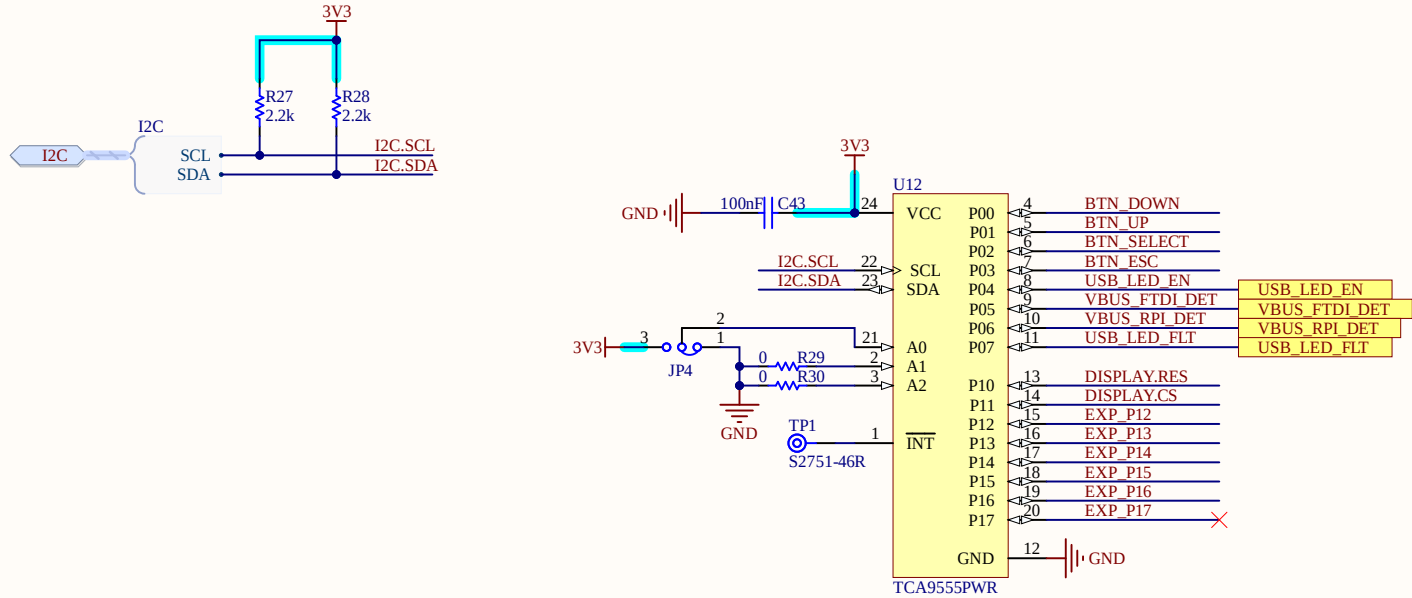
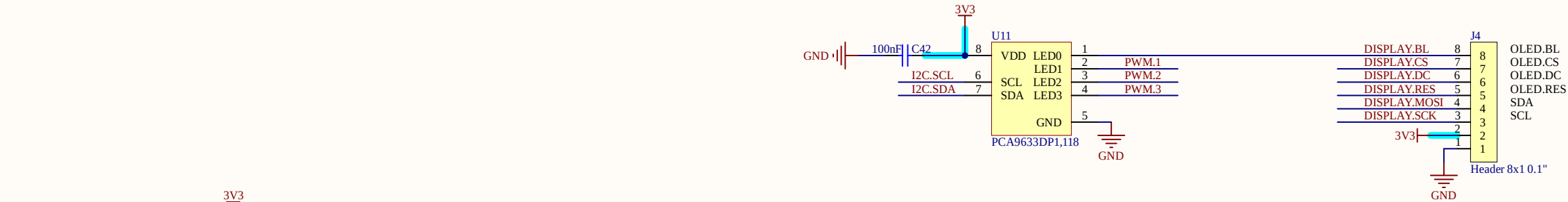
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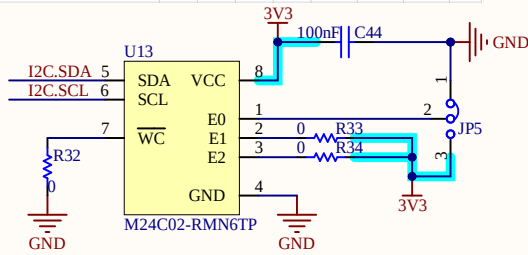
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Package	Device type identifier ⁽¹⁾				Chip Enable address			RW
	b7	b6	b5	b4	b3	b2	b1	
TSSOP8,S08N,UFDFPN6	1	0	1	0	E2	E1	E0	R/W



7-bit I2C address 0x50 through 0x57
Default to 0x56

Title		
Size	Number	Revision
B		
Date:	9/04/2026	Sheet of
File:	C:\Users\...\UI.SchDoc	Drawn By: